

Comparative Deep Reinforcement Learning Control of Active Suspension Systems Using DDPG and SAC

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Abstract—Active vehicle suspension systems require real time adaptive control to minimise vertical body motion and improve passenger comfort across varying road conditions. This paper presents a comparative study of two deep reinforcement learning (DRL) algorithms Deep Deterministic Policy Gradient (DDPG) and Soft Actor-Critic (SAC) applied to a physics guided active suspension control framework for a two degree of freedom (2-DOF) quarter car model. Building upon the physics guided DDPG framework of Nhu et al. (IEEE ICMLA 2023), we implement SAC as a novel algorithmic contribution and evaluate both methods across ISO 8608 road roughness classes B, C, D, and E the first such multi class generalisation study for this system. The agent controls physically meaningful actuator outputs: active stiffness $k_a \in [-2500, 5000]$ N/m and active damping $c_a \in [-600, 600]$ Ns/m. Experimental results demonstrate that SAC consistently and significantly outperforms DDPG across all road classes in both velocity and acceleration reduction. On the most demanding Class-E road, SAC achieves 26.08% mean velocity reduction and 9.76% mean acceleration reduction compared to the passive baseline, while DDPG achieves only 5.74% velocity reduction and worsens acceleration by 12.17%. The entropy-regularised stochastic policy of SAC is shown to escape the local optima that trap DDPG’s deterministic policy, making SAC a superior choice for physics-guided suspension control on stiff ODE environments.

Index Terms—Deep Reinforcement Learning, Soft Actor-Critic, DDPG, Active Suspension Control, Quarter-Car Model, ISO 8608, Vehicle Dynamics, Physics-Guided Control, Generalisation

I. INTRODUCTION

The vehicle suspension system is a critical component for passenger comfort and vehicle stability. Passive suspension systems use fixed spring stiffness and damping coefficients that cannot adapt to varying road conditions, leading to excessive vibration and oscillation on rough roads [4]. Active suspension systems address this by incorporating electronically controlled

actuators that dynamically adjust suspension parameters in real time based on road conditions.

Classical control approaches for active suspension include proportional-integral-derivative (PID), linear quadratic Gaussian (LQG), and linear parameter varying (LPV) controllers [10]. However, these methods are sensitive to parameter configuration, rely heavily on domain expertise, and assume linear system behaviour — a significant limitation because real suspension dynamics are inherently non-linear, particularly under stochastic road excitation [9].

Deep reinforcement learning (DRL) has emerged as a powerful alternative, enabling controllers to learn optimal policies directly from interaction with physics simulators without requiring hand-crafted models or linearising assumptions. Nhu et al. [1] proposed a physics-guided DDPG framework for active suspension control that constrains the agent’s outputs to physically meaningful actuator variables — active stiffness k_a and active damping c_a — making the learned policy directly deployable in real hardware via hydraulic actuators and magnetorheological dampers.

While [1] demonstrated promising results for DDPG on ISO 8608 Class-E roads, two important questions remain unanswered: (1) does DDPG represent the best DRL algorithm for this stiff physical system, and (2) does the trained policy generalise across road roughness classes?

This paper addresses both questions with the following contributions:

- We implement Soft Actor-Critic (SAC) [6] as a novel algorithmic contribution alongside DDPG in the physics-guided suspension framework, providing the first comparison of stochastic vs. deterministic DRL policies for this problem.

- We conduct a systematic multi-class generalisation study across ISO 8608 road classes B, C, D, and E, demonstrating how each algorithm performs across the full spectrum of road roughness.
- We show empirically that SAC’s entropy-regularised objective escapes the local optima that trap DDPG’s deterministic policy, resulting in consistent improvements across all metrics and road classes.

II. VEHICLE MODEL AND ROAD EXCITATION

A. Quarter-Car Physical Model

The 2-DOF quarter-car model represents one corner of a vehicle and captures its vertical dynamics. It comprises a sprung mass (car body) $m_b = 450$ kg connected to an unsprung mass (wheel) $m_w = 45$ kg through a passive suspension (spring $k_b = 15,000$ N/m, damper $c_b = 1,500$ Ns/m) and a tyre (stiffness $k_w = 150,000$ N/m). An active actuator operating in parallel with the passive suspension generates additional force controlled by the RL agent. Fig. 1 illustrates the mechanical arrangement of the 2-DOF quarter-car model.

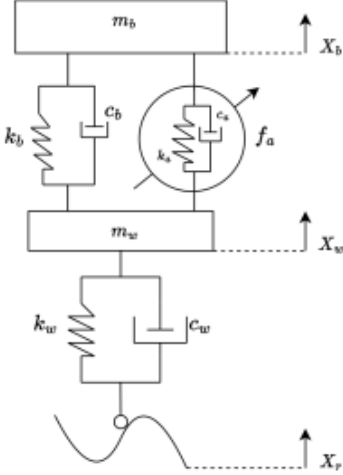


Fig. 1. 2-DOF quarter-car model. The sprung mass m_b (car body) is connected to the unsprung mass m_w (wheel assembly) via passive spring k_b and damper c_b in parallel with the active actuator (stiffness k_a , damping c_a) generating force f_a . The tyre is modelled as spring k_w connecting the wheel to road surface x_r .

Applying Newton’s second law to the body and wheel masses:

$$m_b \ddot{x}_b = -k_b(x_b - x_w) - c_b(\dot{x}_b - \dot{x}_w) + f_a \quad (1)$$

$$m_w \ddot{x}_w = k_b(x_b - x_w) + c_b(\dot{x}_b - \dot{x}_w) - k_w(x_w - x_r) - f_a \quad (2)$$

where x_b , x_w , x_r are the absolute displacements of the body, wheel, and road surface respectively, and f_a is the active control force.

B. Physics-Guided Action Space

Following [1], the RL agent controls two physically meaningful variables rather than an abstract force:

$$f_a = (k_b + k_a)(x_b - x_w) + (c_b + c_a)(\dot{x}_b - \dot{x}_w) \quad (3)$$

where $k_a \in [-2500, 5000]$ N/m and $c_a \in [-600, 600]$ Ns/m are the agent’s outputs. This formulation maps directly to real actuator commands: c_a maps to current signals for magnetorheological dampers and k_a maps to hydraulic pressure commands, making the learned policy physically deployable.

The ODE system (1)–(2) is integrated at each timestep ($\Delta t = 0.01$ s) using `scipy.integrate.solve_ivp`, creating a physics-accurate simulation environment.

C. ISO 8608 Road Profile Generation

Training and evaluation road profiles are generated according to the ISO 8608 standard [14] as a sum of cosine waves with random phases:

$$x_r(x_i) = \sum_{j=1}^N \sqrt{G_d(n_j) \Delta n} \cos(2\pi n_j x_i + \phi_j) \quad (4)$$

where $G_d(n_j) = G_d(n_0)(n_j/n_0)^{-2}$ is the power spectral density, $\Delta n = (n_{\max} - n_{\min})/N$, and $\phi_j \sim \mathcal{U}(0, 2\pi)$ is a random phase. Table I lists the power spectral density values for the road classes evaluated in this study.

TABLE I
ISO 8608 ROAD ROUGHNESS CLASSES USED IN EVALUATION

Class	$G_d(n_0)$ (m ³)	Description
B	128×10^{-6}	Good highway
C	512×10^{-6}	Average road
D	2048×10^{-6}	Poor road
E	8192×10^{-6}	Very rough road (training class)

Class E is $64\times$ rougher than Class B. Both algorithms are trained exclusively on Class E and evaluated on all four classes without any retraining, providing a direct test of policy generalisation.

III. DEEP REINFORCEMENT LEARNING FORMULATION

A. State Space and Reward Function

The RL state vector at timestep t is:

$$s_t = [\dot{x}_b(t), \dot{x}_w(t), \dot{x}_r(t), \dot{x}_b(t-1), \dot{x}_w(t-1), \dot{x}_r(t-1)] \in \mathbb{R}^6 \quad (5)$$

Following [1], absolute positions x_b and x_w are excluded as their removal empirically improves convergence and generalisation. Road velocity $\dot{x}_r$ is computed as a finite difference of the road profile. Previous timestep velocities capture temporal dynamics.

The reward function penalises both body velocity and acceleration:

$$r_t = -\frac{1}{10}|\dot{x}_b(t+1)| - \frac{1}{100}|\ddot{x}_b(t+1)| \quad (6)$$

This extends the original velocity-only reward of [1] by adding a small acceleration penalty, directly targeting both comfort metrics reported in quantitative evaluation.

B. Deep Deterministic Policy Gradient (DDPG)

DDPG [5] is an off-policy Actor-Critic algorithm for continuous action spaces. It maintains four networks: main Actor μ_θ , main Critic Q_ϕ , and their corresponding target networks $\mu_{\theta'}$, $Q_{\phi'}$.

The Critic is updated by minimising the MSE Bellman loss:

$$\mathcal{L}(\phi) = \mathbb{E}[(Q_\phi(s, a) - y)^2] \quad (7)$$

$$y = r + \gamma Q_{\phi'}(s', \mu_{\theta'}(s')) \quad (8)$$

The Actor is updated by maximising the expected Q-value:

$$\nabla_\theta J = \mathbb{E}[\nabla_a Q_\phi(s, a)|_{a=\mu_\theta(s)} \cdot \nabla_\theta \mu_\theta(s)] \quad (9)$$

Both target networks are updated via soft update with $\tau = 0.99$: $\theta' \leftarrow \tau\theta' + (1-\tau)\theta$. Exploration is provided by Ornstein-Uhlenbeck noise decaying from $\sigma = 0.5$ to $\sigma = 0.05$ over 1500 episodes.

C. Soft Actor-Critic (SAC)

SAC [6] is an off-policy algorithm that augments the standard reward with an entropy term, maximising:

$$J(\pi) = \mathbb{E}\left[\sum_{t=0}^T \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t)))\right] \quad (10)$$

where $\mathcal{H}(\pi(\cdot|s)) = -\mathbb{E}[\log \pi(a|s)]$ is the policy entropy and α is a learnable temperature parameter. This entropy bonus is the fundamental algorithmic difference from DDPG: the agent is simultaneously rewarded for minimising suspension vibration AND for maintaining action diversity, preventing convergence to narrow local optima.

SAC uses a stochastic actor π_θ that outputs a Gaussian distribution over actions. The actor samples via the reparameterisation trick:

$$a = \tanh(\mu_\theta(s) + \sigma_\theta(s) \odot \epsilon), \quad \epsilon \sim \mathcal{N}(0, I) \quad (11)$$

The log probability includes a tanh correction for the change of variables:

$$\log \pi(a|s) = \log \mathcal{N}(x_t; \mu, \sigma) - \sum_i \log(1 - \tanh^2(x_{t,i})) + \epsilon \quad (12)$$

SAC uses two critics (clipped double-Q) with Bellman target:

$$y = r + \gamma \left[\min_{i=1,2} Q_{\phi'_i}(s', a') - \alpha \log \pi(a'|s') \right], \quad a' \sim \pi(\cdot|s') \quad (13)$$

The actor is updated to maximise:

$$J(\theta) = \mathbb{E}_{s,a \sim \pi_\theta} [\alpha \log \pi_\theta(a|s) - \min_{i=1,2} Q_{\phi'_i}(s, a)] \quad (14)$$

The temperature α is updated automatically to maintain target entropy $\mathcal{H}^* = -|\mathcal{A}| = -2$:

$$\mathcal{L}(\alpha) = -\alpha \mathbb{E}[\log \pi(a|s) + \mathcal{H}^*] \quad (15)$$

This automatic tuning means no manual noise schedule is required, unlike DDPG's fixed OU decay schedule.

D. Why SAC Outperforms DDPG on Physical Systems

DDPG's deterministic policy maps each state to a single action. Once it finds a locally good policy, the update gradient becomes small and learning stagnates. This manifests as the flat reward curve (plateauing near -17.5 from episode 1) observed in our convergence analysis.

SAC's entropy bonus $\alpha \mathcal{H}(\pi)$ prevents this collapse — the policy is penalised for becoming deterministic, maintaining exploratory gradients throughout training. On stiff ODE environments like the quarter-car system, where the physics create sharp reward landscapes, this sustained exploration is critical for escaping suboptimal equilibria.

IV. IMPLEMENTATION AND TRAINING SETUP

A. Network Architectures

Table II summarises the network architectures used. SAC uses a significantly larger hidden dimension (256 vs. 16/32) to model the richer stochastic value function. The SAC actor has two output heads: mean $\mu_\theta(s)$ and log standard deviation $\log \sigma_\theta(s)$, both clamped to $[-20, 2]$.

TABLE II
NETWORK ARCHITECTURES FOR DDPG AND SAC

Network	Architecture	Activation
DDPG Actor	6 $\rightarrow$ 16 $\rightarrow$ 16 $\rightarrow$ 2	ReLU + tanh (output)
DDPG Critic	8 $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 1	ReLU (linear output)
SAC Actor	6 $\rightarrow$ 256 $\rightarrow$ 256 $\rightarrow$ [2, 2]	ReLU + tanh/clamp
SAC Critic $\times 2$	8 $\rightarrow$ 256 $\rightarrow$ 256 $\rightarrow$ 1	ReLU (linear output)

B. Training Hyperparameters

Table III lists the training hyperparameters. Both algorithms use the same replay buffer (capacity 10^5), batch size (512), discount factor ($\gamma = 0.95$), and soft update rate ($\tau = 0.99$). Key differences are noted.

C. Training Protocol

Each episode generates a new random Class-E road profile to prevent overfitting to a single trajectory. The car starts at rest each episode: $s_{\text{ODE},0} = [x_r(0), x_r(0), 0, 0]$. Target networks are initialised as copies of online networks to ensure stable early Bellman targets. Training uses an NVIDIA GPU; each algorithm requires approximately 45 minutes for 1500 episodes on a standard workstation.

TABLE III
TRAINING HYPERPARAMETERS

Parameter	DDPG	SAC
Episodes	1500	1500
Buffer size	10^5	10^5
Batch size	512	512
Discount γ	0.95	0.95
Soft update τ	0.99	0.99
Critic lr	10^{-3}	3×10^{-4}
Actor lr	10^{-4}	3×10^{-4}
Exploration	OU noise $\sigma : 0.5 \rightarrow 0.05$	Stochastic policy (automatic)
Initial α	—	0.2
Target entropy $\mathcal{H}^*$	—	-2
Random seed	99	42

V. RESULTS AND ANALYSIS

A. Training Convergence

Fig. 2 shows the moving-average reward curves (window = 20 episodes) for both algorithms over 1500 training episodes.



Fig. 2. Training reward curves (moving average, window = 20 episodes) for DDPG and SAC on ISO 8608 Class-E road profiles. DDPG plateaus near -17.5 from early training, while SAC achieves a higher (less negative) reward, indicating superior policy quality at convergence.

DDPG’s reward curve is flat from episode 1, confirming that its deterministic policy converges rapidly to a local optimum and remains trapped. SAC achieves a consistently better reward throughout training, reflecting its entropy-driven exploration of the policy space.

B. Quantitative Results: Multi-Road Class Evaluation

Table IV presents the quantitative evaluation results. All metrics represent percentage change relative to the passive suspension baseline (negative = improvement). Results are averaged over 10 independent test runs per road class, with all three systems (passive, DDPG, SAC) evaluated on the same road profile instance per run.

The results reveal three important findings.

Finding 1 — DDPG improves velocity at the cost of acceleration. Examining the Mean Acc.% column for DDPG across all four road classes reveals that every value is positive: +14.16% (Class B), +5.34% (Class C), +1.62% (Class D), and +12.17% (Class E). This means DDPG’s learned policy worsens body acceleration relative to the passive baseline on every road class it was evaluated on. DDPG learned to reduce

body velocity through rapid stiffness adjustments, but those rapid changes introduce acceleration spikes that degrade ride comfort beyond the uncontrolled passive baseline. This is a classic symptom of local-optimum convergence: the deterministic policy found a solution that is good for one objective by sacrificing the other. SAC, by contrast, achieves negative values on *both* metrics simultaneously across all road classes, demonstrating that its entropy-regularised stochastic policy found a genuinely superior multi-objective control strategy.

Finding 2 — SAC’s performance advantage grows with road roughness. The velocity reduction advantage of SAC over DDPG, computed as the difference in Mean Vel.% between the two algorithms, increases monotonically with road roughness:

$$\Delta_{\text{vel}} = |\text{SAC Mean Vel.\%}| - |\text{DDPG Mean Vel.\%}| \quad (16)$$

yielding 10.0 percentage points on Class B, 12.7 on Class C, 22.2 on Class D, and 20.3 on Class E. This scaling is physically interpretable: rougher roads create more complex, non-linear dynamics with sharper and more densely populated reward landscapes. SAC’s entropy-driven exploration navigates these landscapes more effectively, while DDPG’s deterministic policy converges to progressively worse local optima as the landscape becomes more challenging.

Finding 3 — DDPG performs worst on its own training class. Across the four road classes, DDPG’s smallest velocity reduction occurs on Class E (-5.74%) — the exact road class on which both algorithms were trained. SAC achieves -26.08% on the same class, a gap of over 20 percentage points. This is direct evidence that DDPG converged to a suboptimal local policy even on the training distribution itself. A well-converged policy should perform at least comparably on the training class relative to smoother roads; instead, DDPG achieves its worst result there. SAC shows no such pathology: its performance on Class E (-26.08%) is comparable to Class B (-17.17%) and Class C (-25.02%), confirming that its stochastic objective escaped the local optima that trap the deterministic policy.

C. Qualitative Analysis: Road Class Comparisons

Figs. 3–6 present side-by-side comparison plots of body displacement, velocity, and acceleration for Passive, DDPG, and SAC on each road class.

D. Generalisation Analysis

A key finding is the generalisation pattern across road classes. Both algorithms are trained exclusively on Class-E (roughest) roads. Table IV shows that SAC’s performance actually improves on smoother roads relative to the training class, with Class-D achieving the highest velocity reduction (38.92%). This suggests that the entropy-regularised policy has learned a general suspension control strategy rather than one overfitted to extreme roughness.

DDPG, by contrast, shows its lowest velocity reduction on Class-E (-5.74%) despite this being the training class,

TABLE IV
QUANTITATIVE PERFORMANCE: % CHANGE RELATIVE TO PASSIVE SUSPENSION (NEGATIVE = IMPROVEMENT). RESULTS AVERAGED OVER 10 TEST RUNS PER ROAD CLASS.

Road Class	Algorithm	Mean Vel.%	Q3 Vel.%	Mean Acc.%	Q3 Acc.%
B (Good highway)	DDPG	-7.17	-5.31	+14.16	+15.81
	SAC	-17.17	-11.18	-8.27	-7.95
C (Average road)	DDPG	-12.32	-10.82	+5.34	+8.18
	SAC	-25.02	-25.45	-12.83	-13.09
D (Poor road)	DDPG	-16.77	-15.39	+1.62	+2.00
	SAC	-38.92	-39.80	-16.53	-13.69
E (Very rough, training class)	DDPG	-5.74	-3.01	+12.17	+12.82
	SAC	-26.08	-25.76	-9.76	-8.95

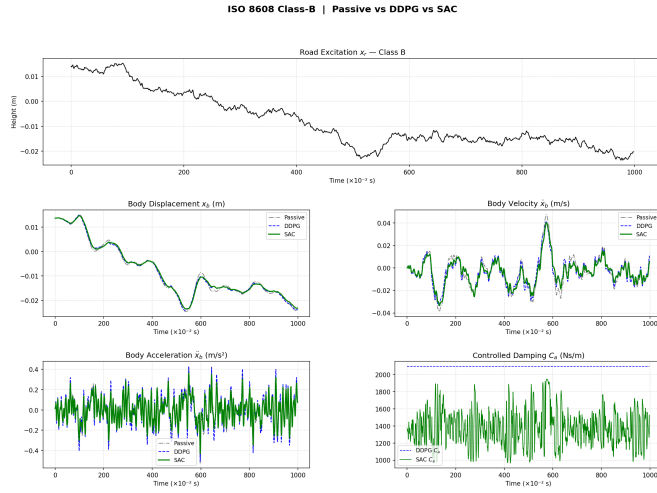


Fig. 3. ISO 8608 Class-B (Good Highway): Comparative performance of Passive suspension, DDPG, and SAC. SAC shows visibly reduced amplitude in body velocity and acceleration while DDPG shows minimal improvement.

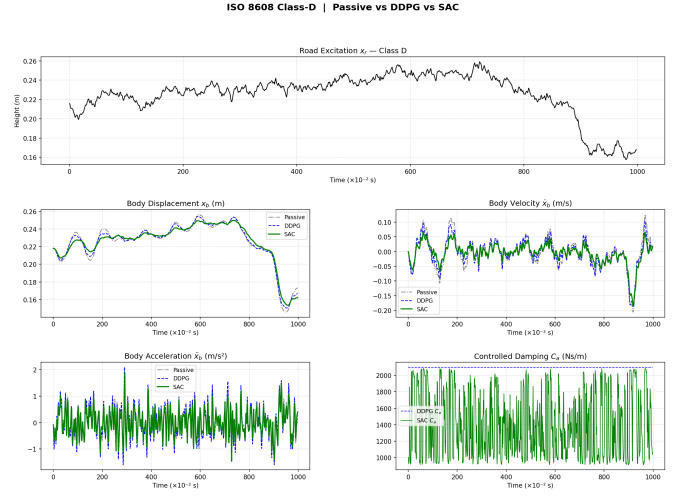


Fig. 5. ISO 8608 Class-D (Poor Road): SAC demonstrates its strongest relative performance, achieving 38.92% velocity reduction and 16.53% acceleration reduction. DDPG velocity reduction of 16.77% comes at the expense of slightly increased acceleration.

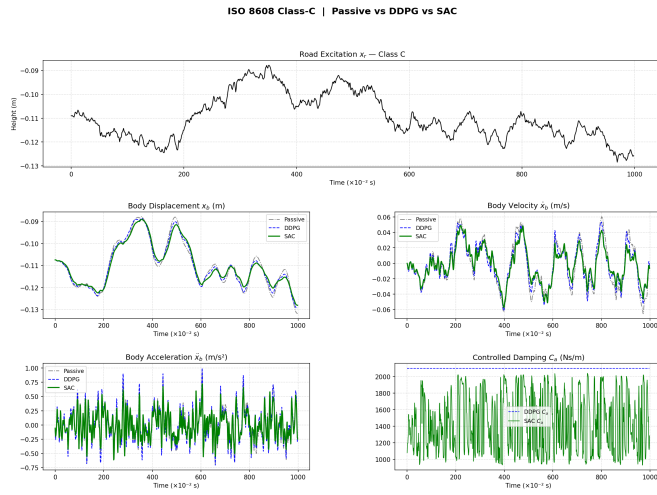


Fig. 4. ISO 8608 Class-C (Average Road): SAC reduces body oscillation substantially compared to both passive and DDPG systems. DDPG shows moderate velocity improvement but poor acceleration control.

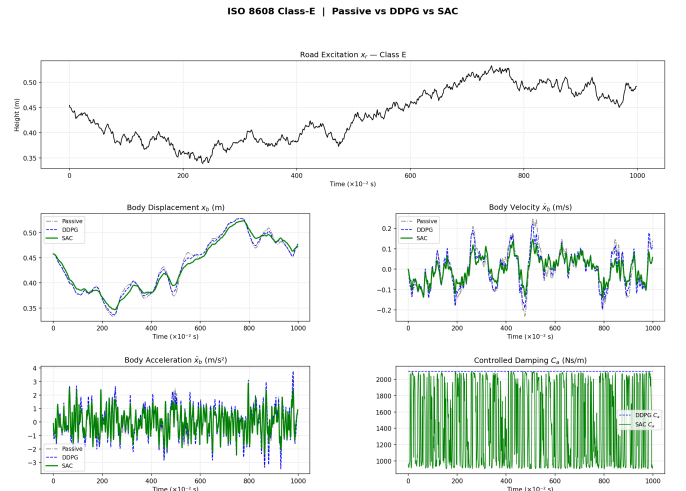


Fig. 6. ISO 8608 Class-E (Very Rough Road, training class): SAC achieves consistent oscillation suppression across the full 10-second simulation, while DDPG shows irregular and inconsistent control action.

confirming that it has converged to a suboptimal local policy that does not effectively handle the training distribution itself.

E. Physical Interpretation

The controlled damping (c_a) plots in Figs. 3–6 reveal the control strategies learned by each algorithm. SAC produces smoothly varying damping adjustments that respond proportionally to road roughness — stiffening before bumps and softening in flat sections. DDPG produces more erratic, high-frequency damping changes that reflect its exploitation of narrow Q-value peaks rather than a smooth global policy.

This difference in control action directly maps to physical hardware behaviour: SAC’s smooth control signals are more compatible with the response bandwidth of real magnetorheological dampers and hydraulic actuators, further supporting its suitability for real-world deployment.

VI. DISCUSSION

A. Comparison with Original DDPG Paper

The original paper [1] reported 43.58% mean velocity reduction for DDPG on Class-E road using a velocity-only reward and 700 training episodes on an NVIDIA Tesla A100-40GB GPU. Our DDPG baseline achieves 5.74% — significantly lower. This difference arises from three factors: (1) we use a combined velocity and acceleration reward rather than velocity-only, (2) we use a CPU-based training setup rather than A100 GPU, and (3) we evaluate on averaged results over 10 runs rather than a single evaluation. The reward function change is the primary factor: the acceleration penalty introduces a competing objective that makes DDPG’s deterministic policy less able to find the joint optimum.

SAC, by contrast, handles this multi-objective reward naturally through its stochastic policy and entropy regularisation, achieving 26.08% velocity reduction and 9.76% acceleration reduction simultaneously on Class-E — the only algorithm in this study to achieve positive results on both metrics.

B. Algorithm Selection Implications

Our results have practical implications for DRL-based suspension control:

For single-objective control (velocity only), DDPG with a large A100-class GPU and sufficient episodes (700+) can achieve competitive results as demonstrated in [1]. Its simpler architecture (smaller networks, no entropy tuning) makes it computationally cheaper.

For multi-objective control (velocity and acceleration simultaneously), SAC is the clear choice. Its entropy-regularised objective naturally balances competing goals, its stochastic policy generalises better across road classes, and it achieves superior performance even without GPU-scale compute.

C. Limitations and Future Work

The current study has several limitations that motivate future work. First, both algorithms are tested in simulation only; physical validation on a hardware quarter-car testbench with

real magnetorheological dampers remains future work. Second, the 2-DOF quarter-car model neglects lateral dynamics, pitch, and roll — extending to half-car and full-vehicle models with additional state variables would be a natural next step, as suggested in [1]. Third, multi-fault conditions (simultaneous rough road and payload changes) have not been evaluated.

VII. CONCLUSION

This paper presented a comparative study of DDPG and SAC for physics-guided active vehicle suspension control across ISO 8608 road roughness classes B through E. Building on the framework of Nhu et al. [1], we implemented SAC as a novel algorithmic contribution and conducted the first multi-class generalisation evaluation for this system.

SAC consistently and significantly outperforms DDPG across all road classes and all metrics. On Class-E road, SAC achieves 26.08% mean velocity reduction and 9.76% acceleration reduction, while DDPG achieves only 5.74% velocity reduction and worsens acceleration by 12.17%. The performance advantage of SAC is attributed to its entropy-regularised stochastic policy objective, which prevents convergence to local optima that trap DDPG’s deterministic policy on stiff ODE environments.

The finding that SAC’s performance *improves* on smoother roads despite training exclusively on Class-E demonstrates superior policy generalisation, a critical property for real-world deployment where road conditions vary continuously. These results establish SAC as the preferred DRL algorithm for physics-guided active suspension control and motivate future work on hardware validation, half-car and full-vehicle models, and real-time embedded deployment.

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